Module – 1
Introduction to Cyber Security

Content

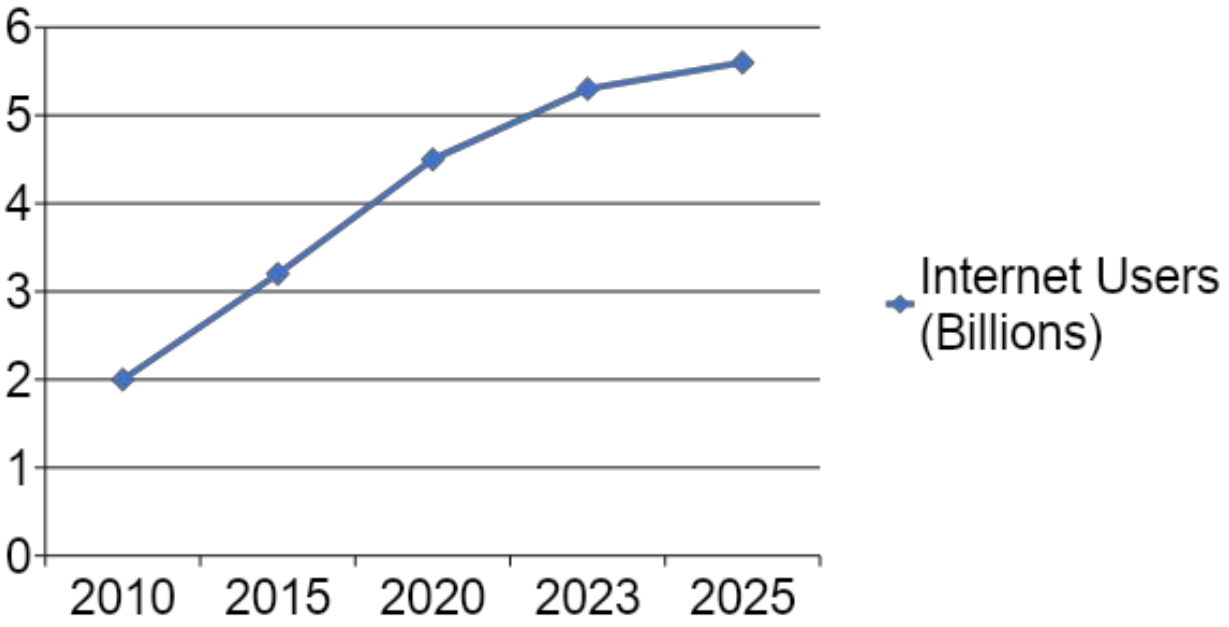
- Importance of Cyber Security in the Digital World
- Key Terminology and Basic Concepts
- Principles of Information Security (Confidentiality, Integrity, Availability)
- Evolution and Trends in Cyber Security

Importance of Cyber Security in the Digital World

Introduction to the Digital World :

- 5+ Billion Internet Users Worldwide

  - Online Banking & UPI Payments
  - Social Media Platforms
  - Cloud Storage & IoT Devices
  - Every activity leaves a digital footprint



Importance of Cyber Security in the Digital World

Internet users worldwide — last 5 years (annual snapshot)

Year	Internet users (approx.)
2021	4.9 – 5.0 billion.
2022	~5.0 – 5.35 billion.
2023	5.35 billion (DataReportal 2024 global overview).
2024	~5.35–5.6 billion (annual growth slowed but still positive).
2025	~6.0 billion (ITU estimate for 2025 / ~74% global penetration).

Importance of Cyber Security in the Digital World

Global active social media users (total, all platforms combined):

Year	Active social media users (global, approx.)
2021	4.20 billion.
2022	4.62 billion.
2023	4.76 billion.
2024	5.04 billion.
2025	5.24 billion (Backlinko / DataReportal aggregates).

Importance of Cyber Security in the Digital World

Global active social media users (total, all platforms combined):

Facebook (Meta) — ~3.0+ billion monthly active users (MAU) regionally aggregated across Facebook products.

YouTube — ~2+ billion logged-in users per month

WhatsApp / Instagram (Meta) — 2+ billion (WhatsApp) and ~2+ billion (Instagram + combined family

TikTok — ~1.0–1.5 billion MAU (rapid growth 2020–2024).

Increase in cyber-crime — Global picture (cost / trend)

- Industry projection (Cybersecurity Ventures): **global cybercrime costs** forecast at **~$9.5 trillion (2024)** and **~$10.5 trillion (2025)** (loss/damage estimates including fraud, lost productivity, restoration, etc.)
- Cybercrime estimated at **$3 trillion in 2015**, rising quickly over the decade to the multi-trillion range.
- Costs are rising year-on-year, driven by larger-scale ransomware, supply-chain attacks, and more monetized fraud operations. Expect continued upward pressure as attackers adopt automation and AI

2018 – $4 Trillion
2021 – $6 Trillion
2023 – $8 Trillion
2025 – $10 Trillion (Projected)

Increase in cyber-crime — India

- CERT-In reported incidents handled in 2023: 1,592,917 incidents
- CERT-In reported incidents handled in 2025: ~2,944,000 (29.44 lakh) incidents

Growth of the cyber-security industry — global & India

- global end-user spending on information security projected at ~$212 billion in 2025
- India Cybersecurity Domestic Market 2023: India cybersecurity market reached ~USD 6.06 billion in 2023

Career Opportunities

- Ethical Hacker
- SOC Analyst
- Cyber Forensic Expert
- Security Engineer
- Chief Information Security Officer (CISO)

Key Terminologies in Cyber Security

Basic Security Concepts

- Confidentiality – Ensuring information is accessible only to authorized users.
- Integrity – Ensuring data is accurate and not altered without authorization.
- Availability – Ensuring systems and data are accessible when needed.
- Authentication – Verifying the identity of a user or system.
- Authorization – Granting access rights after authentication.
- Accountability – Tracking user actions within a system.

Key Terminologies in Cyber Security

Threat-Related Terminologies

- Threat – A potential cause of harm to a system or data.
- Vulnerability – A weakness in a system that can be exploited.
- Exploit – A method used to take advantage of a vulnerability.
- Attack – An attempt to gain unauthorized access or damage a system.
- Risk – The potential loss when a threat exploits a vulnerability.

Key Terminologies in Cyber Security

Malware Terminology

- Malware – Malicious software designed to harm systems.
- Trojan – Malware disguised as legitimate software.
- Worm – Self-replicating malware that spreads automatically.
- Virus – Malware that attaches itself to legitimate programs.
- Ransomware – Malware that encrypts data and demands payment.
- Spyware – Software that secretly monitors user activity.

Key Terminologies in Cyber Security

Social Engineering Terms

- Phishing – Fraudulent attempt to steal sensitive information via fake emails/websites.
- Smishing – Phishing via SMS.
- Vishing – Phishing via phone calls.
- Social Engineering – Manipulating people to reveal confidential information.

Key Terminologies in Cyber Security

Network Security Terms

- Firewall – Monitors and controls network traffic.
- IDS (Intrusion Detection System) – Detects suspicious activities.
- IPS (Intrusion Prevention System) – Prevents detected threats.
- VPN (Virtual Private Network) – Secure encrypted internet connection.
- DDoS – Overloading a server to make it unavailable.

Key Terminologies in Cyber Security

Modern Security Concepts

- Cloud Security – Protecting cloud-based systems and data.
- Endpoint Security – Protecting end-user devices.
- Zero Trust – Never trust, always verify.
- AI in Cyber Security – Using artificial intelligence to detect threats.

Key Terminologies in Cyber Security

Security Operations Terms

- SOC – Security Operations Center.
- Penetration Testing – Simulated cyber attack to test security.
- Vulnerability Assessment – Identifying weaknesses.
- Incident Response – Handling cyber attacks.
- Digital Forensics – Investigating cyber crimes.

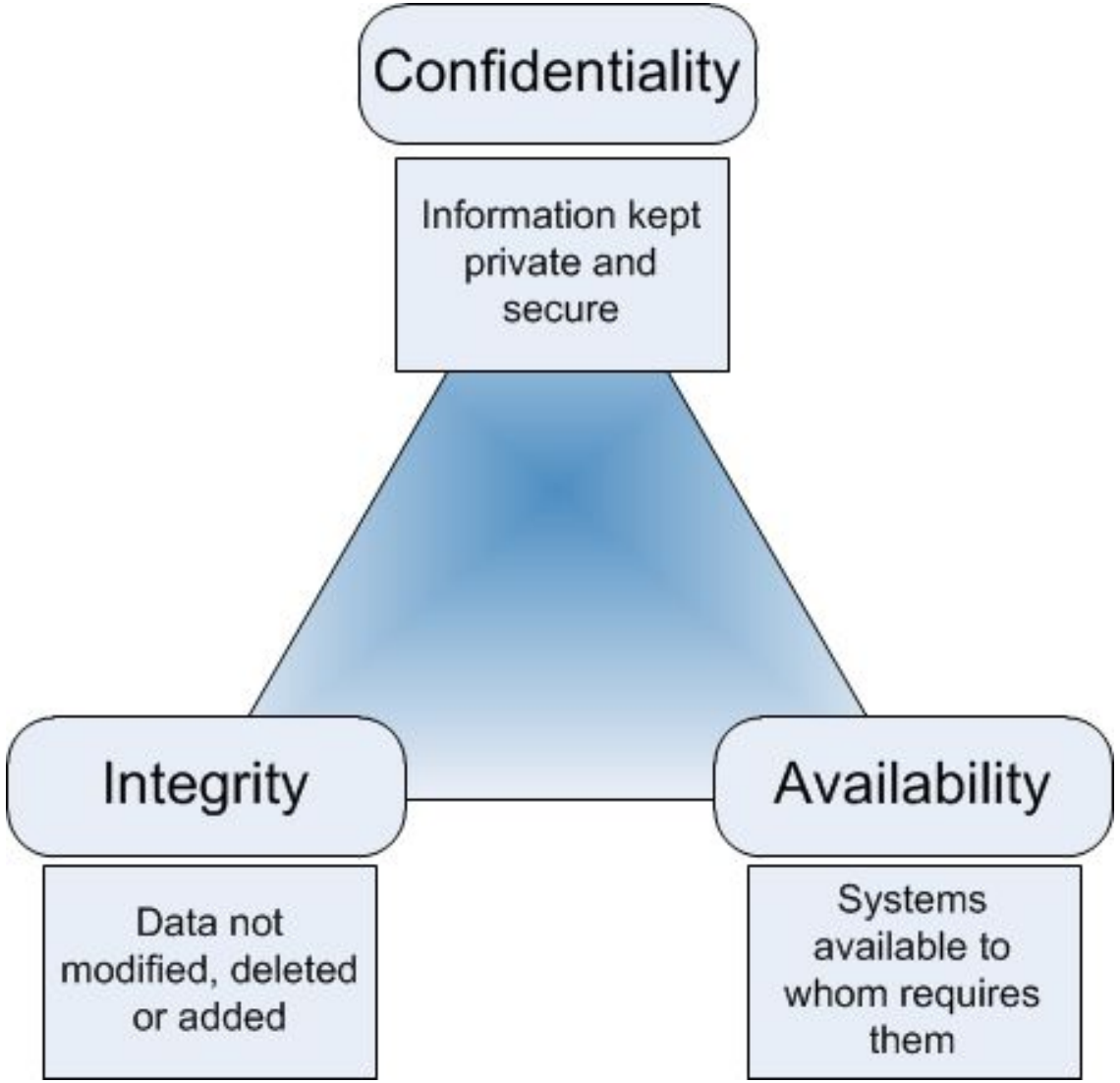
Cyber Security

- **Cybersecurity** is the body of technologies, processes and practices designed to protect networks, computers, programs and data from attack, damage or unauthorized access. In a computing context, **security** includes both **cybersecurity** and physical **security**.

History of Hacking and Cyber Security

- The origin of cyber security began with a research project. It only came into existence because of the development of viruses.
- In 1970's, Robert (Bob) Thomas who was a researcher for BBN Technologies in Cambridge, created the first computer worm (virus).
- He realized that it was possible for a computer program to move across a network, leaving a small trail (series of signs) wherever it went.
- He named the program Creeper, and designed it to travel between Tenex terminals on the early ARPANET, printing the message "I'M THE CREEPER: CATCH ME IF YOU CAN."

CIA


Confidentiality
Information kept private and secure
Integrity
Data not modified, deleted or added
Availability
Systems available to whom requires them

Confidentiality

- *Confidentiality* refers to protecting information from unauthorized access. In other words, only the people who are authorized to do so can gain access to sensitive data.
- A loss of confidentiality is the unauthorized disclosure of information.
- Examples of information that could be considered confidential are health records, financial account information, criminal records, source code, trade secrets, and military tactical plans.
- Security controls that can provide confidentiality protection are encryption, logical and physical access controls, database views, controlled traffic flows, etc.

Integrity

- *Integrity* refers to ensuring the authenticity of information—that information is not altered, and that the source of the information is genuine.
- A loss of integrity is the unauthorized modification or destruction of information.
- One example of information integrity is when a user issues a request to her online bank account to pay her $50 water utility bill. The bank needs to be sure that the integrity of that transaction was not altered during transmission, so the user does not end up paying the utility company $500 instead.
- Security controls that can provide integrity protection are message digest, digital signature, logical and physical access controls, backups, file integrity monitoring, database views, etc.

Availability

- *Availability* means that information should be available when authorized users need to access it.
- A loss of availability is the disruption of access to or use of information or an information system.
- Administrators experience the loss of availability when a file server goes offline or a highly used database is out of service for one reason or another.
- Security controls that can provide availability protection include load balancing, clustering, controlled traffic flows, intrusion detection, fault tolerance and recovery mechanisms, etc.

Hacking

- Hacking is the act of finding the possible entry points that exist in a computer system or a computer network and finally entering into them.
- Hacking is usually done to gain unauthorized access to a computer system or a computer network, either to harm the systems or to steal sensitive information available on the computer.

Types of hacking

- **Website Hacking:** Hacking a website means taking unauthorized control over a web server and its associated software such as databases and other interfaces.
- **Network Hacking:** Hacking a network means gathering information about a network by using tools like Telnet, NS lookup, Ping, Tracert, Netstat, etc. with the intent to harm the network system and hamper its operation.
- **Email Hacking:** It includes getting unauthorized access on an Email account and using it without taking the consent of its owner.

Types of hacking

- **Ethical Hacking:** Ethical hacking involves finding weaknesses in a computer or network system for testing purpose and finally getting them fixed.
- **Password Hacking:** This is the process of recovering secret passwords from data that has been stored in or transmitted by a computer system.
- **Computer Hacking:** This is the process of stealing computer ID and password by applying hacking methods and getting unauthorized access to a computer system.

Purpose of Hacking

- There could be various positive and negative intentions behind performing hacking activities. Here is a list of some probable reasons why people indulge in hacking activities:
- Just for fun
- Show-off
- Steal important information
- Damaging the system
- Money extortion
- System security testing
- To break policy compliance

Advantages of Hacking

- To recover lost information, especially in case you lost your password.
- To perform penetration testing to strengthen computer and network security.
- To put adequate preventative measures in place to prevent security breaches.
- To have a computer system that prevents malicious hackers from gaining access.

Disadvantages of Hacking

- Massive security breach.
- Unauthorized system access on private information.
- Privacy violation.
- Denial of service attacks
- Malicious attack on the system.

Hackers

- A Hacker is a person with a strong interest in computers who enjoys learning and experimenting with them. Hackers are usually very talented, smart people who someone who breaks into computers others.
- The term is often confused with cracker that defines someone who break into computers.

Cracker

- A cracker is a person who break into computer. Crackers should not be confused with hackers. The term “crackers” is usually connected to computer criminals. Some of their crimes include vandalism, theft and snooping in unauthorized areas.

Types of Hacker

- White hat hacker
- Black hat hacker
- Gray hat hacker

White hat hacker

- White Hat hackers are also known as Ethical Hackers. They never intent to harm a system, rather they try to find out weaknesses in a computer or a network system as a part of penetration testing and vulnerability assessments.

- Ethical hacking is not illegal and it is one of the demanding jobs available in the IT industry. There are numerous companies that hire ethical hackers for penetration testing and vulnerability assessments.

Black Hat Hackers

- Black Hat hackers, also known as **crackers**, are those who hack in order to gain unauthorized access to a system and harm its operations or steal sensitive information.

- Black Hat hacking is always illegal because of its bad intent which includes stealing corporate data, violating privacy, damaging the system, blocking network communication, etc.

Grey Hat Hackers

- Grey hat hackers are a blend of both black hat and white hat hackers. They act without malicious intent but for their fun, they exploit a security weakness in a computer system or network without the owner's permission or knowledge.

- Their intent is to bring the weakness to the attention of the owners and getting appreciation or a little bounty from the owners.

Miscellaneous Hackers

- Apart from the above well-known classes of hackers, we have the following categories of hackers based on what they hack and how they do it:
- **Red Hat Hackers**
- **Blue Hat Hackers**
- **Elite Hackers**
- **Script Kiddie**
- **Neophyte/Newbie**
- **Hacktivist**

Type of Hackers

- **Red Hat Hackers**
- Red hat hackers are again a blend of both black hat and white hat hackers. They are usually on the level of hacking government agencies, top-secret information hubs, and generally anything that falls under the category of sensitive information.

- **Blue Hat Hackers**
- A blue hat hacker is someone outside computer security consulting firms who is used to bug-test a system prior to its launch. They look for loopholes that can be exploited and try to close these gaps. Microsoft also uses the term **BlueHat** to represent a series of security briefing events.

Type of Hackers

- **Elite Hackers**
- This is a social status among hackers, which is used to describe the most skilled. Newly discovered exploits will circulate among these hackers.

- **Script Kiddie**
- A script kiddie is a non-expert who breaks into computer systems by using pre-packaged automated tools written by others, usually with little understanding of the underlying concept, hence the term **Kiddie .**

Type of Hackers

- **Neophyte**
- A neophyte, "n00b", or "newbie" or "Green Hat Hacker" is someone who is new to hacking or phreaking and has almost no knowledge or experience of the workings of technology and hacking.

- **Hacktivist**
- A hacktivist is a hacker who utilizes technology to announce a social, ideological, religious, or political message. In general, most hacktivism involves website defacement or denial-of-service attacks.

Vulnerability

- A vulnerability is a weakness that can be exploited by cybercriminals to gain unauthorized access to a computer system. After exploiting a vulnerability, a cyberattack can run malicious code, install malware, and even steal sensitive data.

- Example:
  - Weak password
  - Software Bugs
  - Computer Virus

THREAT

- **Any potential danger to information or systems.**

Risk

- **Using Vulnerability, if someone explore threat then it is known as risk.**

Cyber Attacks

- In computers and computer networks an attack is any attempt to expose, alter, disable, destroy, steal or gain unauthorized access to or make unauthorized use of an Asset.

- A **cyber attack** is any type of offensive action that targets computer information systems, infrastructures, computer networks or personal computer devices, using various methods to steal, alter or destroy data or information systems.

List of Cyber Attacks

- Denial-of-service (DoS) and distributed denial-of-service (DDoS) attacks
- Man-in-the-middle (MitM) attack
- Phishing attacks
- Password attack
- SQL injection attack
- Cross-site scripting (XSS) attack
- Eavesdropping attack
- Birthday attack
- Malware attack

Denial-of-service (DoS) and Distributed denial-of-service (DDoS) attacks

- A denial-of-service attack overwhelms a system's resources so that it cannot respond to service requests. A DDoS attack is also an attack on system's resources, but it is launched from a large number of other host machines that are infected by malicious software controlled by the attacker.

- Denial-of-service doesn't provide direct benefits for attackers. For some of them, it's enough to have the satisfaction of service denial.

Denial-of-service (DoS) and Distributed denial-of-service (DDoS) attacks

- If the attacked resource belongs to a business competitor, then the benefit to the attacker may be real enough.
- Another purpose of a DoS attack can be to take a system offline so that a different kind of attack can be launched. One common example is session hijacking.
- There are different types of DoS and DDoS attacks; the most common are TCP SYN flood attack, teardrop attack, ping-of-death attack and botnets.

Virus (Vital Information Resource Under Slege)

- Computer virus is a program that can "infect" legitimate programs by modifying them to include a possibly "evolved" copy of itself.
- Viruses spread themselves, without the knowledge or permission of the users, to potentially large numbers of programs on many machines.
- A computer virus passes from computer to computer in a similar manner as a biological virus passes from person to person.

Virus

- Viruses may also contain malicious instructions that may cause damage, the combination of possibly Malicious Code with the ability to spread is what makes viruses a considerable concern.

- Viruses can often spread without any readily visible symptoms. A virus can start on event-driven effects(e.g., triggered after a specific number of executions), time-driven effects (e.g. triggered on a specific date, such as Friday the 13th) or can occur at random.

Virus

- Viruses can take some typical action:
  - Display a message to prompt an action which may set of the virus;
  - Delete files inside the system into which viruses enter;
  - Scramble data on a hard disk;
  - Cause erratic screen behavior;
  - Halt the system (PC);
  - Just replicate themselves to propagate further harm.

Worm

- A computer worm is a standalone malware computer program that replicates itself in order to spread to other computers.
- Often, it uses a computer network to spread itself, relying on security failures on the target computer to access it.
- Worms almost always cause at least some harm to the network, even if only by consuming bandwidth, whereas viruses almost always corrupt or modify files on a targeted computer.

Worm

- The **Morris worm** or **Internet worm** of November 2, 1988, was one of the first computer worms distributed via the Internet, and the first to gain significant mainstream media attention.
- It also resulted in the first felony conviction in the US under the 1986 Computer Fraud and Abuse Act.
- It was written by a graduate student at Cornell University, Robert Tappan Morris, and launched on November 2, 1988, from the computer systems of the MIT.

Basis for Comparison	Virus	Worms
Meaning	The Virus attaches itself to executable files and transfers from one system to the other.	A Worm is a malicious program that replicates itself and can spread to different computers via Network.
Human Action	Needed	Not Required
Speed of Spreading	Slower as compared to Worm	Fast
Requirement of host	Host is needed for spreading.	It doesn't need a host to replicate from one computer to another.
Protect the System using	Antivirus software	Antivirus, firewall
Consequences	Corrupt and erase a file or program.	Consumes system resources and slows down it, and can halt the system completely.

Basis for Comparison	Virus	Worms
Different type	Stealth virus, self-modified virus, encryption with variable key virus, polymorphic code virus, metamorphic code virus	E-Mail worms, instant messaging worms, Internet worms, IRC worms, file-sharing network worms
Inception	The creeper virus was considered as the first known virus. It was spread through ARPANET in the early 1970s. It spreads through the TENEX OS and uses connected modem to dial out to a remote computer and infect it.	The name worm originated from The Shockwave Rider, a science fiction novel published in 1975 by John Brunner. Later researchers John F Shock and Jon A Hupp at Xerox PARC published a paper in 1982, The Worm Programs and after that the name was adopted

Trojan Horses

- Trojan Horse is a program in which malicious or harmful code is contained inside apparently harmless programming or data in such a way that it can get control and cause harm, for example, running the file allocation table on the hard disk.

- A Trojan Horse may get widely redistributed as part of a computer virus. The term Trojan Horse comes from Greek mythology about the Trojan War.

Trojan Horses

- Mostly Trojans are introduced via email attachments. These emails are disguised in a way that they look authentic. Once the user downloads the attached file and runs it, the file starts corrupting the system.
- A Trojan can also come as a payload with freeware and shareware available on the Internet. Although every freeware doesn't come with Trojan, it is still advised that one should download software and freeware from authentic sources only.

Trojan Horses

- Trojans can have multiple usages, which depend on the motives of the attacker. These could be identity theft, data theft, crashing computers, spying or tracking user activities.
- They do not replicate but can come attached to a virus which can spread to other computers across the network.

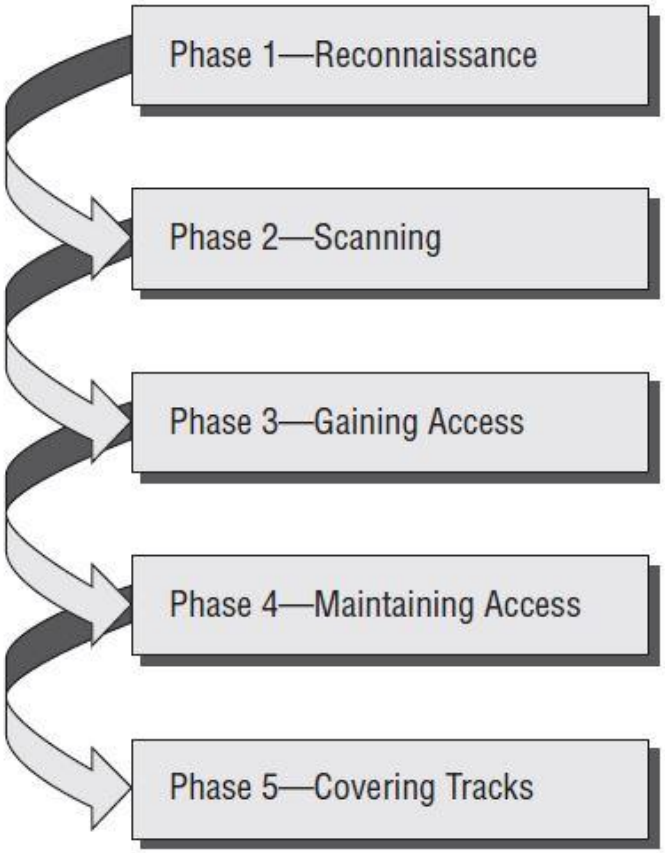
Example of threats by Trojan

1. They erase, overwrite or corrupt data on a computer.
2. They help to spread other malware such as viruses.
3. They deactivate or interfere with antivirus and firewall programs.
4. They upload and download files without your knowledge.
5. They gather E-mail address and use them for spam.

Basis for comparison	Virus	Worm	Trojan Horse
Meaning	A computer program that connects itself to another legitimate program to cause harm to the computer system or the network.	It eats resources of a system to bring it down rather than performing destructive actions.	It permits an intruder to obtain some confidential information about a computer network.
Execution	Depends on the transfer of a file.	Replicates itself without any human action.	Downloaded as software and executed.
Replication occurs	Yes	Yes	No
Remotely controlled	No	Yes	Yes
Rate of spreading	Moderate	Faster	Slow
Infection	Initiates by attaching a virus to an executable file.	Utilizes system or application weaknesses.	Attaches itself to a program and interpret as useful software.
Purpose	Modification of the information.	Halt the CPU and memory.	Steals the user's information.

The Phases of Ethical Hacking

Phases of hacking



Phase 1: Passive and Active Reconnaissance

- *Passive reconnaissance* involves gathering information about a potential target without the targeted individual's or company's knowledge.
- Passive reconnaissance can be as simple as watching a building to identify what time employees enter the building and when they leave. However, most reconnaissance is done sitting in front of a computer.

Phase 1: Passive and Active Reconnaissance

- Active Reconnaissance is a foundational phase in cybersecurity where an attacker, or authorized penetration tester, interacts **directly** with a target system, network, or application to gather detailed, real-time intelligence.
- Unlike passive reconnaissance, which relies on public, non-interactive sources (OSINT), active reconnaissance is intrusive and generates traffic, making it more likely to be detected by security systems like firewalls and Intrusion Detection Systems (IDS).

•**Identifying Live Hosts:** Discovering which IP addresses are active.
•**Port Scanning:** Finding open ports and services running on them.
•**Service Enumeration:** Determining the specific versions of software running to find known exploits.
•**Operating System (OS) Fingerprinting:** Identifying the target's OS to tailor attacks.
•**Mapping Network Topology:** Creating a map of the network infrastructure.

Phase 2: Scanning

- *Scanning* involves taking the information discovered during reconnaissance and using it to examine the network. Tools that a hacker may employ during the scanning phase include
  - Dialers
  - Port scanners
  - Internet Control Message Protocol (ICMP) scanners
  - Ping sweeps
  - Network mappers
  - Simple Network Management Protocol (SNMP) sweepers
  - Vulnerability scanners

Phase 3: Gaining Access

- Once ethical hackers expose vulnerabilities through the process's first and second hacking phases, they now attempt to exploit them for administrative access. The third phase involves attempting to send a malicious payload to the application through the network, an adjacent subnetwork, or physically using a connected computer. Hackers typically use many hacking tools and techniques to simulate attempted unauthorized access.

Phase 4: Maintaining Access

- The fourth phase of the ethical hacking process involves processes to ensure the hacker can access the application for future use. A white-hat hacker continuously exploits the system for further vulnerabilities and escalates privileges to understand how much control attackers can gain once they pass security clearance. Some attackers may also try to hide their identity by removing the evidence of an attack and installing a backdoor for future access.

Phase 5: Covering Track

- To avoid any evidence that leads back to their malicious activity, hackers perform tasks that erase all traces of their actions. These include:
- Uninstalling scripts/applications used to carry out attacks
- Modifying registry values
- Clearing logs
- Deleting folders created during the attack

DDoS

- **Low Orbit Ion Cannon (LOIC)**
- **High Orbit Ion Cannon (HOIC)**
- **Slowloris**
- **R.U.D.Y (R-U-Dead-Yet)**